

Project Design Phase Problem – Solution Fit

Date	01 October 2026
Team ID	SWTID-2026-2391
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why.

Purpose:

- Solve complex problems in a way that fits the state of your customers.
- Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- Sharpen your communication and marketing strategy with the right triggers and messaging.
- Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- **Understand the existing situation in order to improve it for your target group.**

Template:

Problem-Solution Fit canvas for the Script-Controlled ACL project:

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS - ServiceNow administrators and security_admin users who configure access - Institution / department staff (faculty) who manage student records - Students and branch users (ECE, EEE, CSE) who need limited record visibility	6. CUSTOMER CONSTRAINTS CC - Needs security_admin elevation to create ACLs - Limited ServiceNow scripting knowledge - Risk of wrong role assignment exposing data - Cannot change platform ACL engine	5. AVAILABLE SOLUTIONS AS - Default table-level ACLs: role only, no field/branch logic - Manual checking of records: slow and error-prone - Separate tables per branch: duplication, hard to maintain - Pros: simple. Cons: not fine-grained	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P - Restrict who can read records based on branch / field value - Control CREATE, WRITE and DELETE separately per role - Keep full administrative access - Verify access quickly for each user type	9. PROBLEM ROOT CAUSE RC - Role-only ACLs cannot express record-level conditions - Sensitive student data sits in one shared table - No single script-based rule enforcing who sees what - Access rules are not tested by role combination	7. BEHAVIOUR BE - Admins create users, roles and table records - Impersonate users to check results - Add roles bb1-bb4 step by step to test CRUD - Re-check ACLs after every change	
Identify strong TR & EM	3. TRIGGERS TR - Unauthorized user seen opening other branch records - Need to protect student data in a shared table	10. YOUR SOLUTION SL - Custom Institution Details table (u_institution_details) with Branch choice field - Advanced READ ACL script: admin and bb1 allowed, all others denied - Separate role-based ACLs: CREATE=bb2, WRITE=bb3, DELETE=bb4 - Admin keeps full access; behaviour verified via impersonation	8. CHANNELS of BEHAVIOUR CH - 8.1 ONLINE: ServiceNow UI, user impersonation, ACL list, Student Records list	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM - Before: worried, unsure who can see data - After: confident, in control of access		8.2 OFFLINE CH - Team review of test results and documentation - Training sessions for admins and faculty	

Canvas content (editable text version):

Section	Content
1. Customer Segment(s) (CS)	ServiceNow administrators / security administrators who configure access; institution and department staff (faculty) who manage student records; students and branch users (ECE, EEE, CSE) who need limited access.
2. Jobs-to-be-done / Problems (J&P)	Restrict record access using user roles and field values; control READ, CREATE, WRITE and DELETE separately; keep full administrator access; verify access quickly for each type of user.
3. Triggers (TR)	An unauthorized user is able to open records that belong to another branch; need to protect student details held in one shared table.
4. Emotions: Before / After (EM)	Before: worried and unsure about who can see or change student data. After: confident and in control of record access.
5. Available Solutions (AS)	Default table-level role ACLs (role only, no field-based logic); manual checking of records (slow, error-prone); separate tables per branch (duplication and extra maintenance). Pros: simple. Cons: not fine-grained.
6. Customer Constraints (CC)	Elevation to security_admin is required to create ACLs; limited ServiceNow scripting knowledge; wrong role assignment can expose data; the platform ACL engine itself cannot be changed.
7. Behaviour (BE)	Administrators create users, roles and records; impersonate users to check results; add roles bb1–bb4 step by step to test CRUD operations; re-verify ACLs after every change.
8. Channels of Behaviour (CH)	8.1 Online: ServiceNow interface – User Administration, Tables, Access Control (ACL) list, Student Records list, user impersonation. 8.2 Offline: team review of test results, project documentation, training of administrators and faculty.
9. Problem Root Cause (RC)	Role-only ACLs cannot express record-level conditions; sensitive student data is stored in one shared table; there is no single script-based rule deciding who sees what; access rules are often not tested for each role combination.
10. Solution (SL)	A custom Institution Details table (u_institution_details) with a Branch field, an advanced READ ACL script (admin and bb1 allowed, everyone else denied) and separate role-based ACLs: CREATE = bb2, WRITE = bb3, DELETE = bb4. Administrator keeps full access and behaviour is verified using impersonation.

References:

<https://www.ideahackers.network/problem-solution-fit-canvas/>

<https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>